

of an NFC business card because it determines both the reading range and the amount of energy harvested to power the LED. A poorly designed antenna reduces both the communication range and the harvested power.

For this system, the antenna is designed with an inductance of $2.8\mu\text{H}$ - $3.3\mu\text{H}$ (target: $3\mu\text{H}$), making it suitable for the ST25DV64K NFC IC. The antenna resonates at approximately 13.7MHz-14MHz, providing optimum performance at the NFC operating frequency of 13.56MHz.

Several free antenna design tools are available from STMicroelectronics, NXP, and other NFC manufacturers. The antenna for this system was designed using the ST NFC antenna design tool because the system is based on the ST25DV64K NFC IC.

The antenna uses a multi-turn

rectangular spiral coil optimised for a business card. The design consists of 6-7 turns, with outer dimensions of approximately $33\text{-}34\text{mm} \times 33\text{mm}$, a track spacing of 0.29mm , and a copper thickness of $35\mu\text{m}$ (1oz). These dimensions provide a target inductance of approximately $3\mu\text{H}$, making the antenna well suited to the ST25DV64K NFC IC.

EFY note. NFC antenna performance depends on several factors, including PCB layout, track dimensions, copper thickness, the matching network, and surrounding materials. A detailed article covering antenna theory, design, tuning, and free simulation tools will be considered for publication in a future issue.

Antenna design tools.

ST eDesignSuite antenna tool recommended for this system is available at:

<https://eds.st.com/antenna/>

This free online tool allows users to enter antenna dimensions, the number of turns, track width, spacing, copper thickness, and PCB layers to calculate the inductance and optimise the antenna for ST NFC dynamic tags such as the ST25DV64K.

NXP NFC antenna design hub.

<https://www.nxp.com/products/rfid-nfc/nfc-hf/nfc-readers/nfc-antenna-design-hub:NFC-ANTENNA-DESIGN-TOOL>

NXP provides antenna design guides, calculators, and simulation resources that are useful for general NFC antenna design and optimisation, regardless of the NFC chip used. Fig. 4 shows the NFC antenna design tool.

PCB design

The PCB is a two-layer design, with all SMD components mounted on the top layer. The bottom silkscreen layer can be customised with a logo, QR code, and other branding or design elements. Similarly, the top silkscreen layer can include the system name, custom graphics, QR codes, power specifications, CPU information, contact details, or any other required text. Fig. 5 shows the PCB layout developed for the NFC business card.

SMD work

Fig. 6 shows the custom-designed NFC-enabled electronic business card PCB during the SMD assembly stage. The PCB includes two printed NFC antennas, the NFC IC and its supporting SMD components, a QR code, and personalised details, illustrating how a conventional business card can be transformed into a smart digital contact-sharing device.

The SMD components are assembled on the PCB. Assembly can be carried out in two ways: by using a PCB stencil supplied by the PCB manufacturer for accurate solder paste application or by manually applying solder paste directly to the component pads for small-scale prototyping. After the components are positioned, they are soldered using a reflow oven or a hot-air rework station. Once soldering is complete, the joints should be inspected and any excess flux removed. The hardware

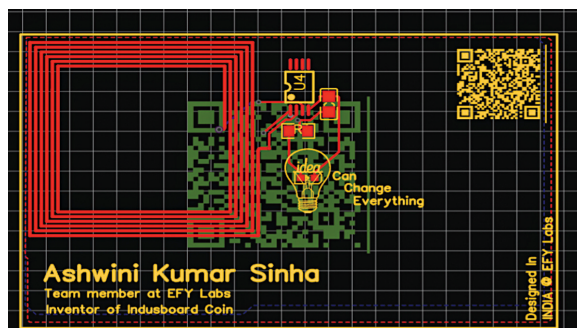


Fig. 5: PCB of NFC business card



Fig. 6: SMD soldering work on NFC business card